

Chapter 27

Bacteria and Archaea

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

Neil Campbell and Jane Reece

Lectures by Chris Romero, updated by Erin Barley with contributions from Joan Sharp

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings

Overview: Masters of Adaptation

- Prokaryotes thrive almost everywhere, including places too acidic, salty, cold, or hot for most other organisms.
- There are more prokaryotes in a handful of fertile soil than the number of people who have ever lived.
- They have an astonishing genetic diversity.
- **Prokaryotes** are divided into **two domains: bacteria and archaea.**

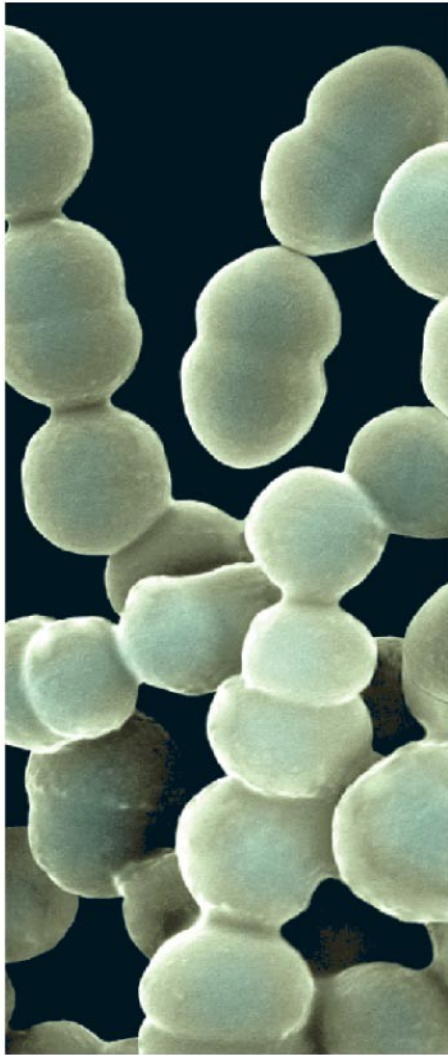
Why is this lakebed red?



Concept 27.1: Structural and functional adaptations contribute to prokaryotic success

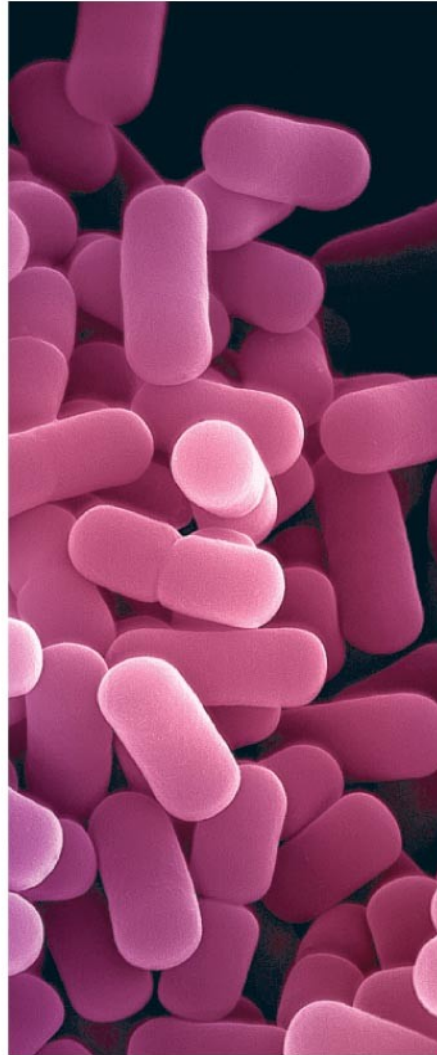
- Most prokaryotes are **unicellular**, although some species form colonies.
- Most prokaryotic cells are 0.5–5 μm , **much smaller** than the 10–100 μm of many eukaryotic cells.
- Prokaryotic cells *three most common shapes are **spheres (cocci)**, **rods (bacilli)**, and **spirals**.*

Prokaryotic Shapes



1 μm

**(a) Spherical
(cocci)**



2 μm

**(b) Rod-shaped
(bacilli)**



5 μm

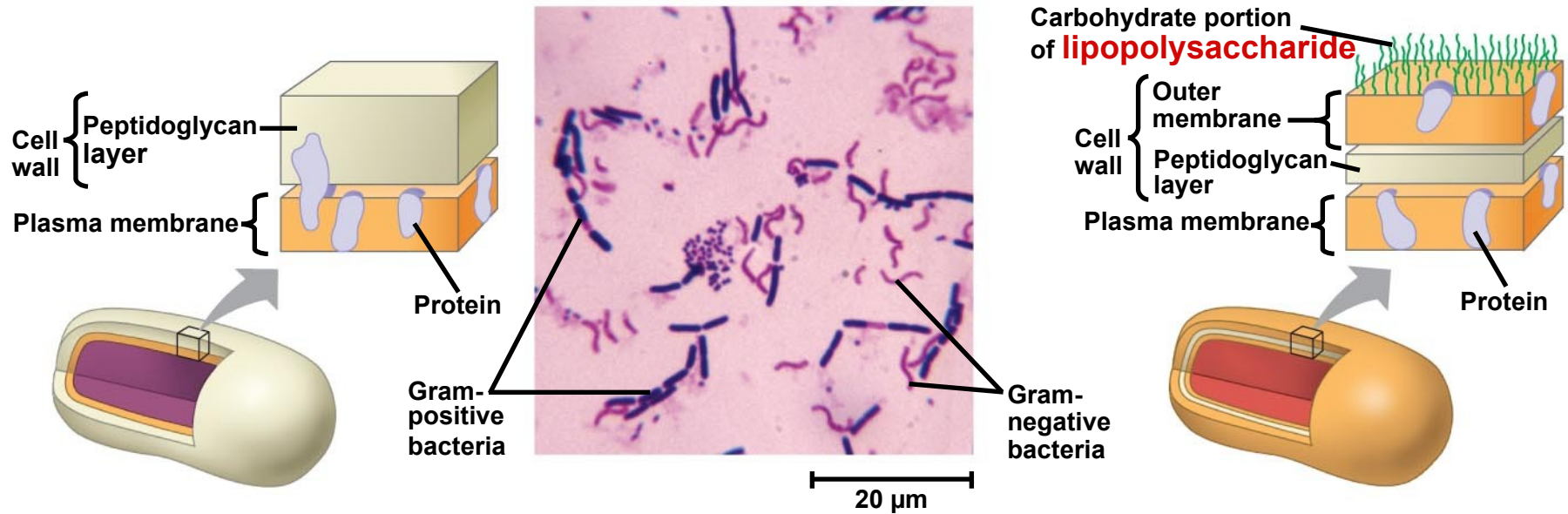
(c) Spiral

Prokaryotes Cell-Surface Structures

- An important feature of nearly all prokaryotic cells is their **cell wall**, which maintains cell shape, provides physical protection, and prevents the cell from bursting in a hypotonic environment.
- Eukaryote cell walls are made of cellulose or chitin.
- **Bacterial cell walls** contain **peptidoglycan**, a network of **sugar polymers** cross-linked by **polypeptides**.

-
- Using the **Gram stain**, scientists classify many bacterial species into **Gram-positive** and **Gram-negative** groups based on cell wall composition.
 - *Gram-negative bacteria* have less peptidoglycan and an *outer membrane* that can be *toxic*, and they are more likely to be *antibiotic resistant*.
 - Many antibiotics target peptidoglycan and damage bacterial cell walls... (often causing cell lysis).

Bacteria: Cell Wall Composition: Gram Stain



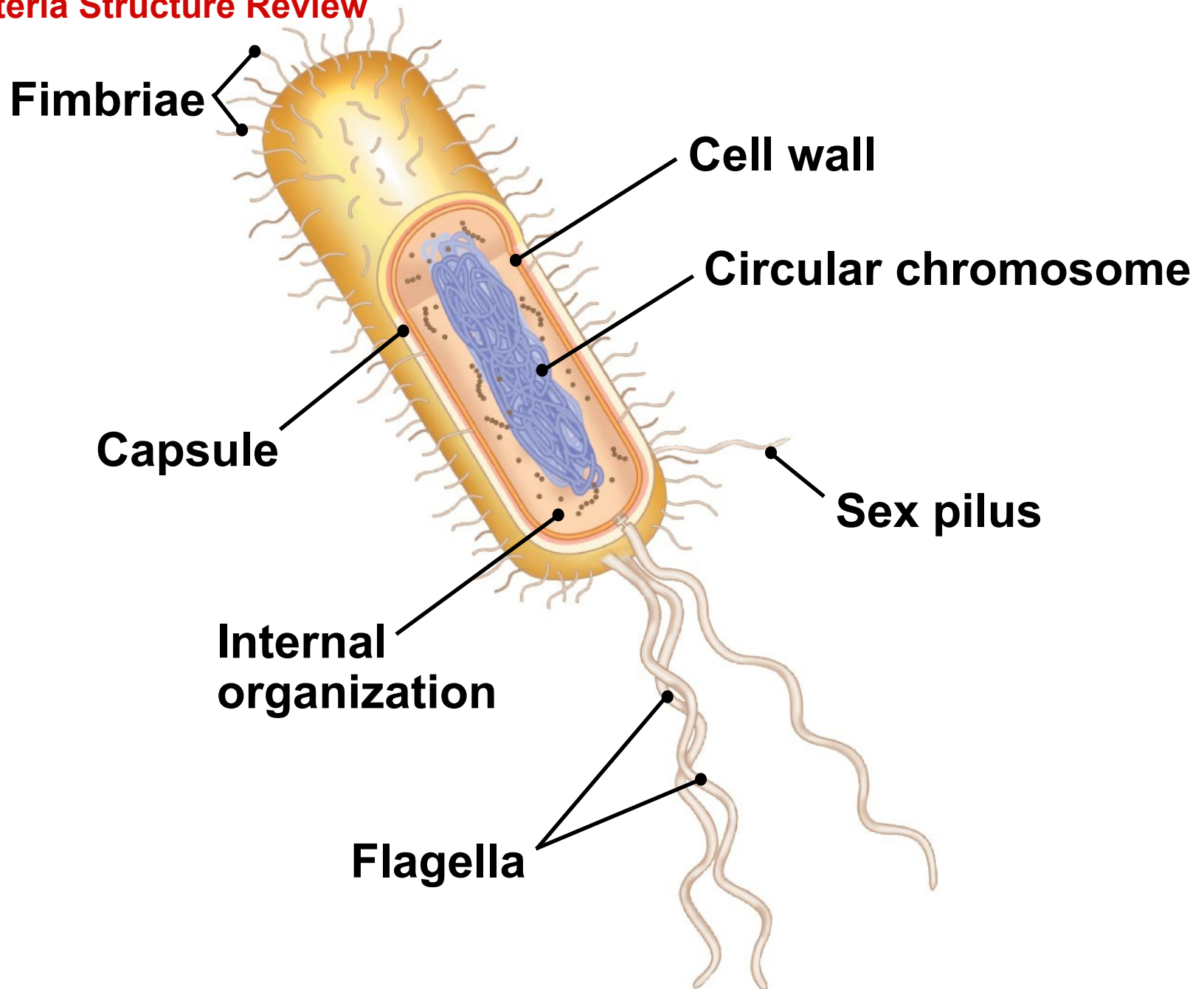
(a) **Gram-positive**: peptidoglycan traps crystal violet.

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

(b) **Gram-negative**: crystal violet is easily rinsed away, revealing red dye.
toxic bacteria

-
- A polysaccharide or protein layer called a **capsule** covers many prokaryotes.
 - Some prokaryotes have **fimbriae** (also called *attachment pili*), which allow them to stick to their substrate or other individuals in a colony
 - **Sex pili** are longer than fimbriae and allow prokaryotes to **exchange DNA** by **conjugation**.

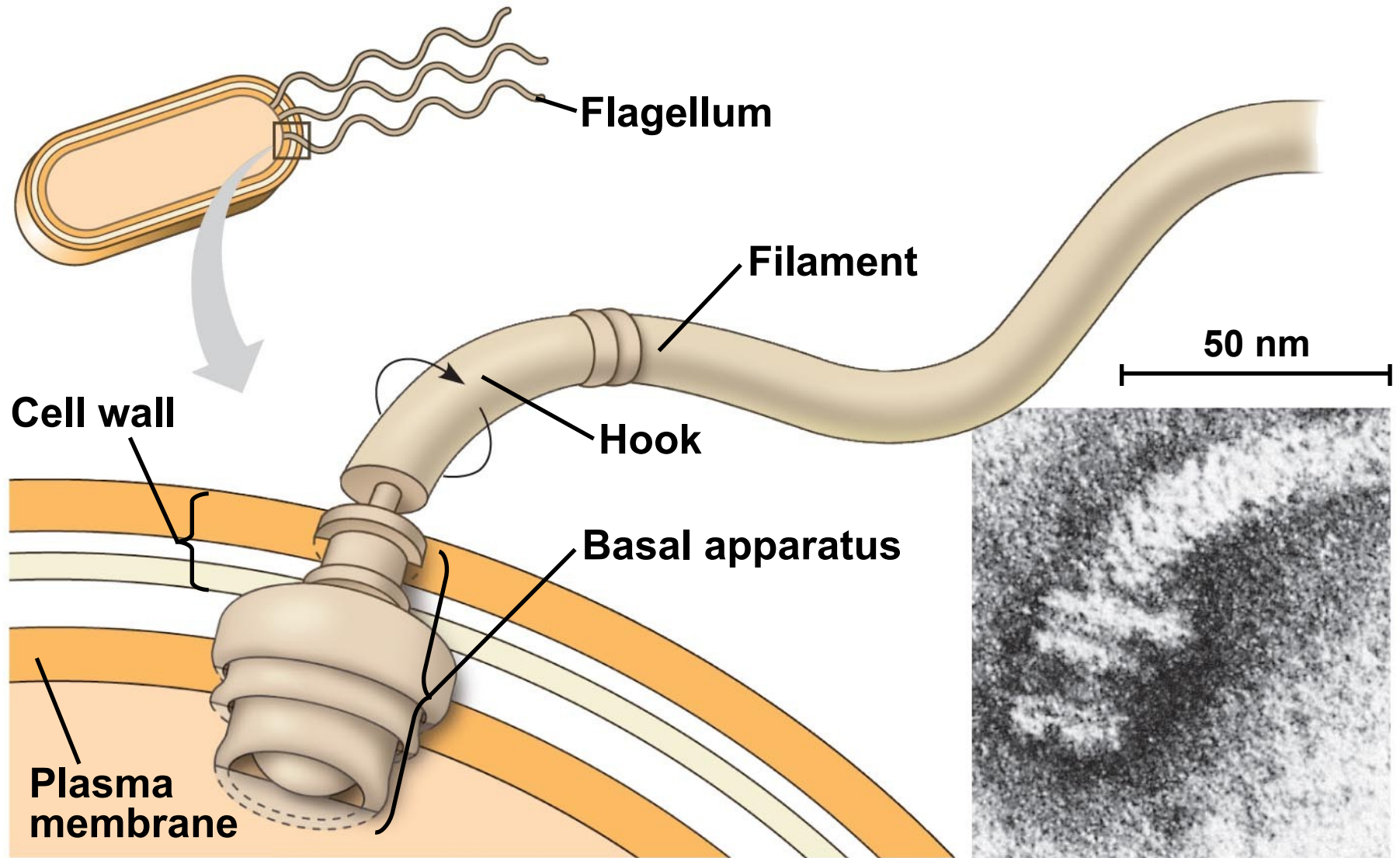
Bacteria Structure Review



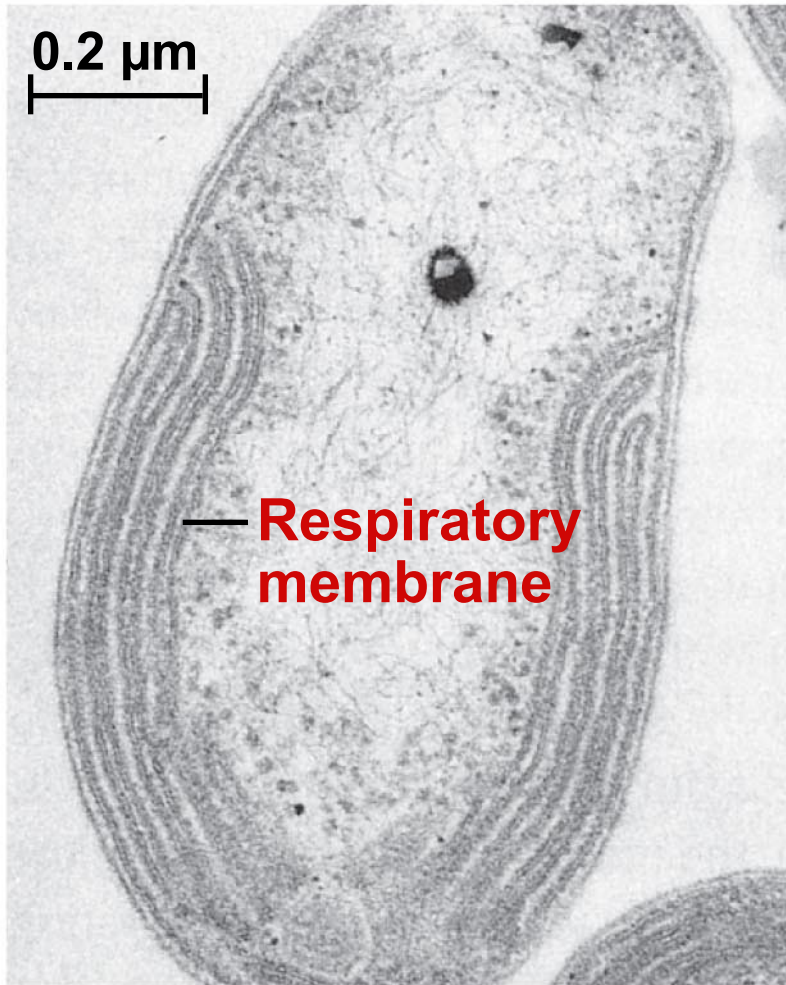
Motility / Internal Organization

- Most **motile bacteria** propel themselves by **flagella** that are structurally and functionally different from eukaryotic flagella.
- In a heterogeneous environment, many bacteria exhibit **taxis**, the *ability to move toward or away from certain stimuli*.
- Prokaryotic cells usually **lack** complex **compartmentalization**.
- Some prokaryotes do have **specialized membranes** that perform **metabolic functions**.

Bacteria flagellum - Structurally different from Eukaryotic flagella



Bacteria often have **Specialized Internal Membranes**



(a) Aerobic prokaryote

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.



(b) Photosynthetic prokaryote

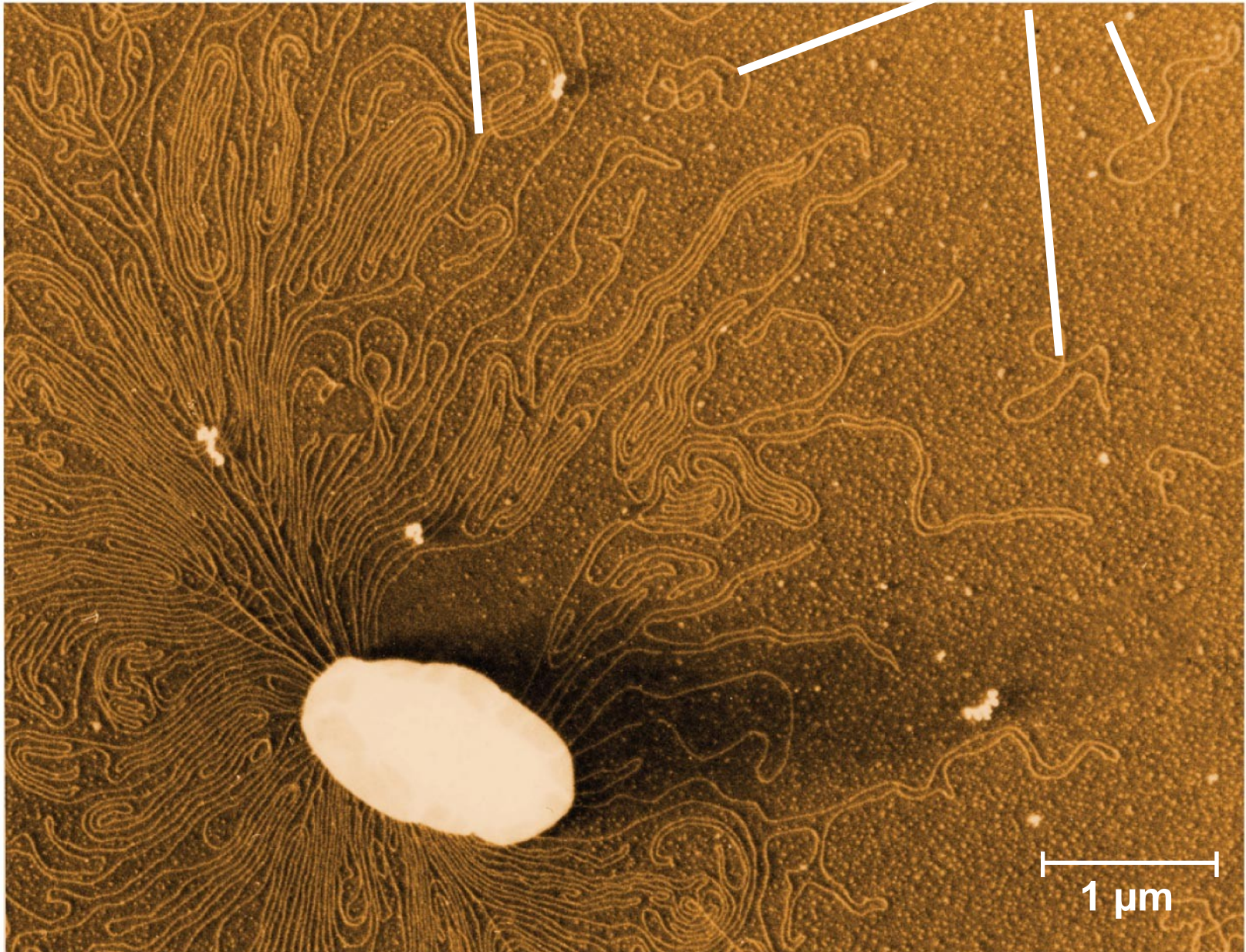
Prokaryotic Genome - No Nucleus

- The **prokaryotic genome** has less DNA than the eukaryotic genome.
- Most of the genome consists of a ***circular chromosome***.
- Some species of bacteria also have **smaller rings of DNA** called **plasmids**.
- The typical prokaryotic genome is a ring of **DNA** that is not surrounded by a membrane and that is located in a **nucleoid region**.

Single Circular **Chromosome**

Plasmids

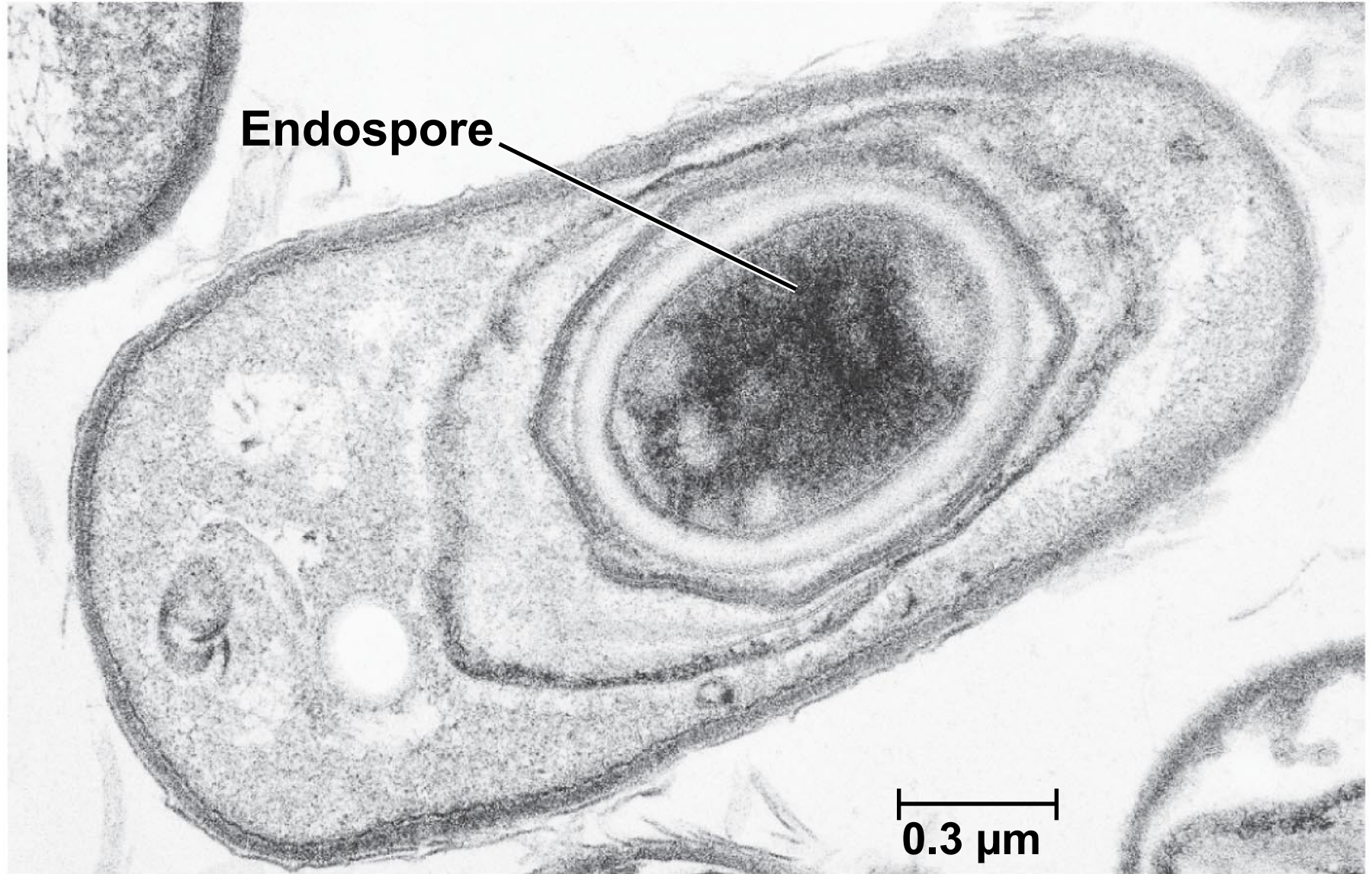
smaller DNA rings



Reproduction and Endospores Adaptation

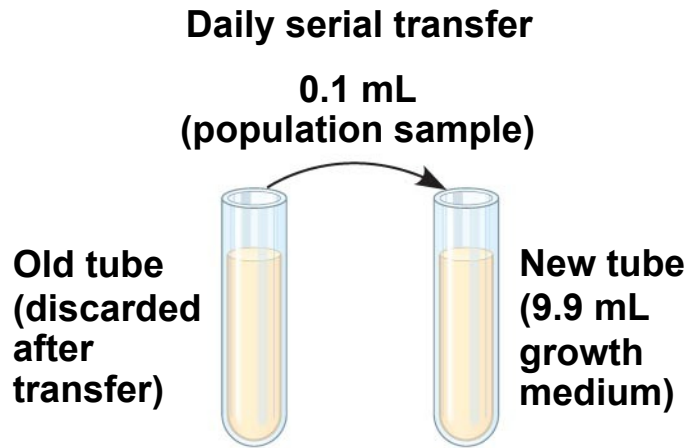
- Prokaryotes reproduce **asexually** quickly by **binary fission** and can **clone** / divide every 1–3 hours.
- Many prokaryotes form metabolically inactive **endospores**, which can remain **dormant** and viable in harsh conditions for centuries.
- Prokaryotes can evolve rapidly because of their short generation times.

Endospores = Adaptation to Survive Dormant in Harsh Environments

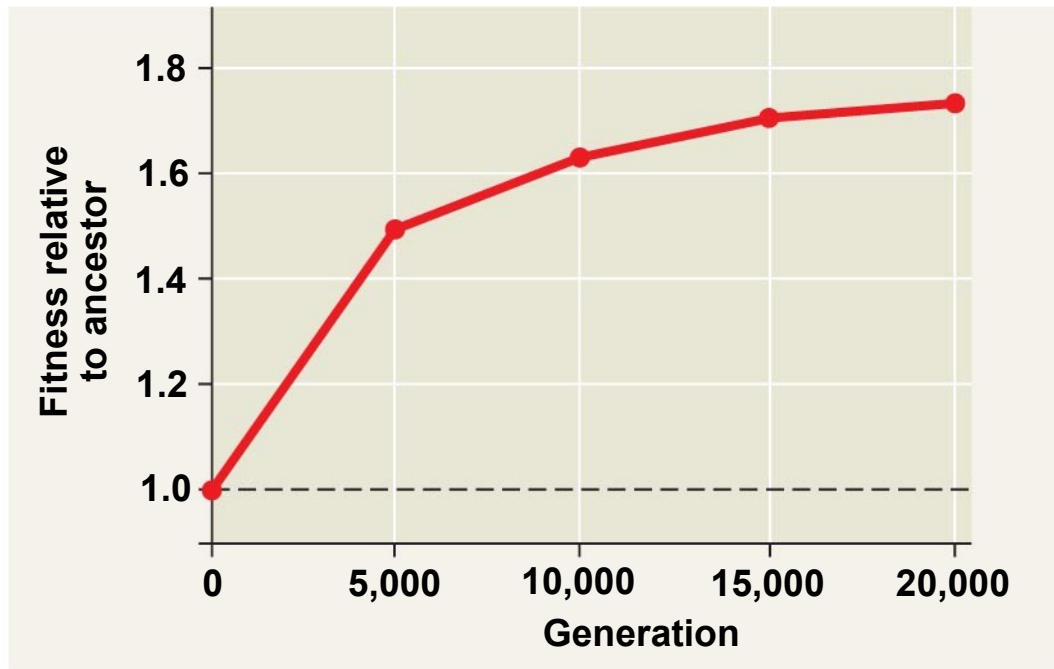


Bacteria Evolve Rapidly in Response to Environmental Conditions

EXPERIMENT



RESULTS



Concept 27.2: Rapid reproduction, mutation, and genetic recombination promote **genetic diversity** in **prokaryotes**

- Prokaryotes have considerable genetic variation.
- **Three factors** contribute to this genetic diversity:
 - ***Rapid reproduction***
 - ***Mutation***
 - ***Genetic recombination***

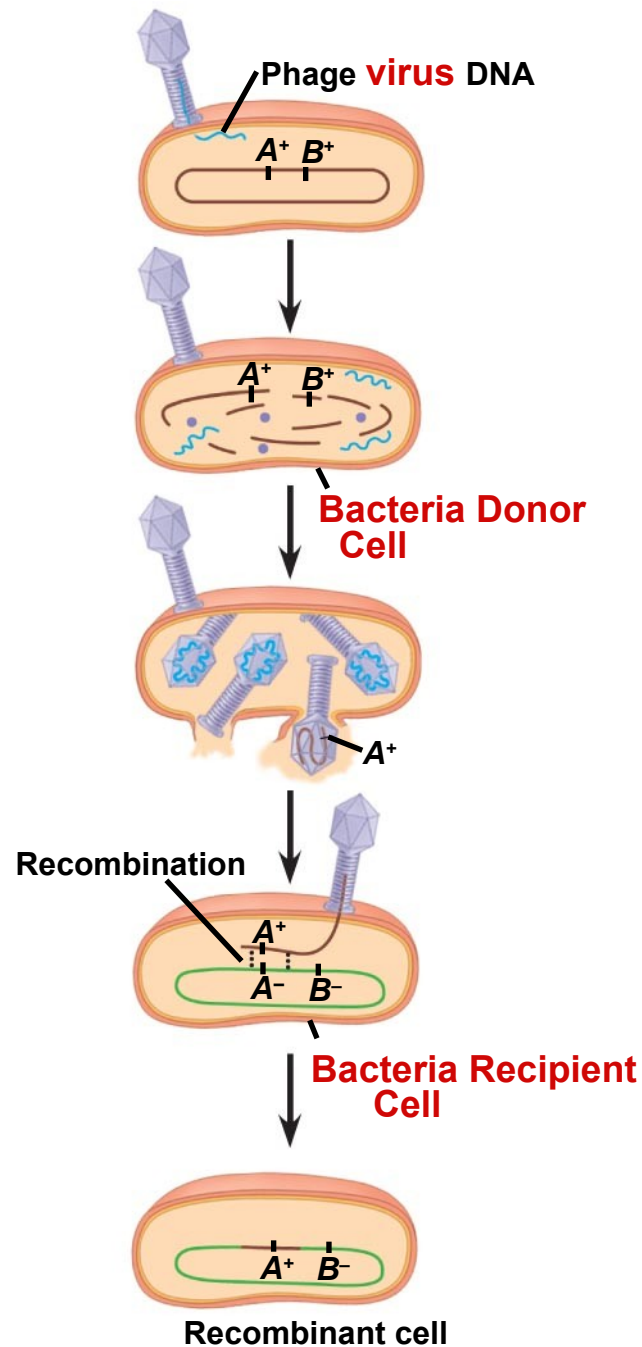
Rapid Reproduction and Mutation

- Prokaryotes reproduce by binary fission, and offspring cells are generally identical.
- ***Mutation*** rates during **binary fission** are **low**, **but** because of ***rapid reproduction***, mutations can **accumulate rapidly** in a population.
- High diversity from mutations allows for ***rapid evolution***.

Genetic Recombination

- Prokaryotic DNA from different individuals can be brought together by ***transformation***, ***transduction***, and ***conjugation***.
- A prokaryotic cell can take up and incorporate ***foreign DNA from*** the surrounding ***environment*** in a process called ***transformation***.
- ***Transduction*** is the ***movement of genes between bacteria by*** bacteriophages (***viruses*** that infect bacteria)

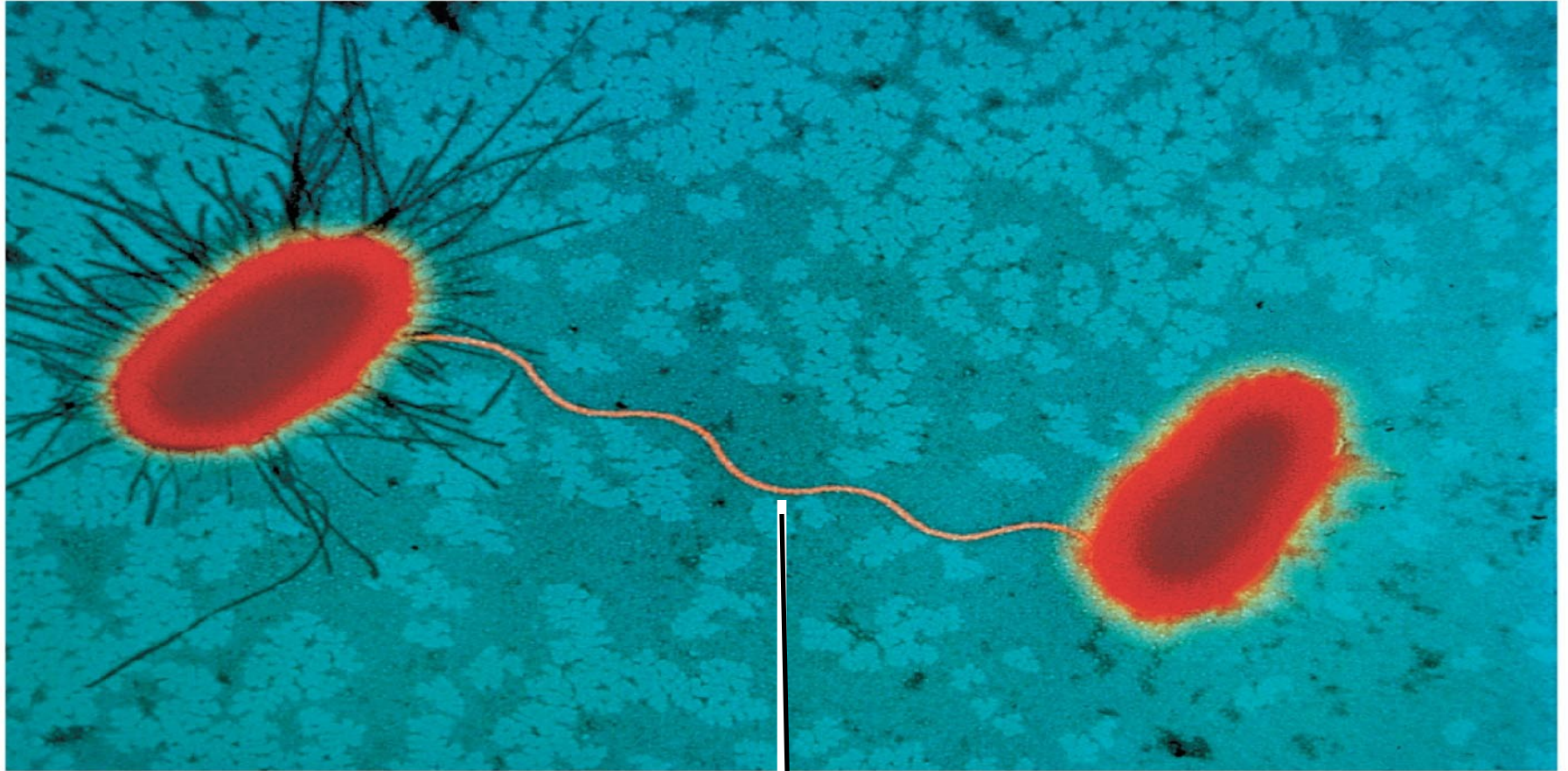
Transduction



Conjugation and Plasmids

- **Conjugation** is the process where *genetic material is transferred between bacterial cells*.
- **Sex pili** allow cells to connect and pull together for *DNA transfer*.
- A piece of DNA called the **F factor** is required for the *production of sex pili*.
- The F factor can exist as a separate plasmid or as DNA within the bacterial chromosome and is transferable during conjugation.

Conjugation

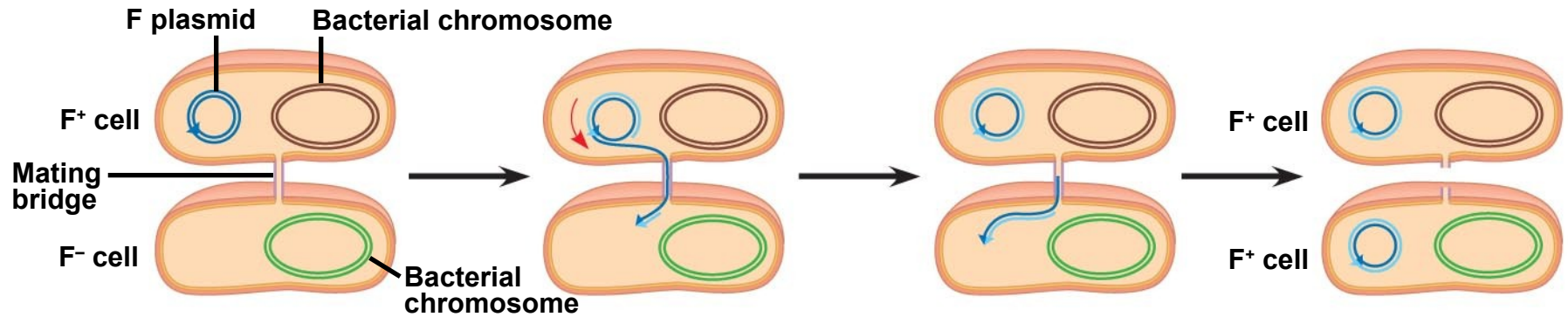


Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

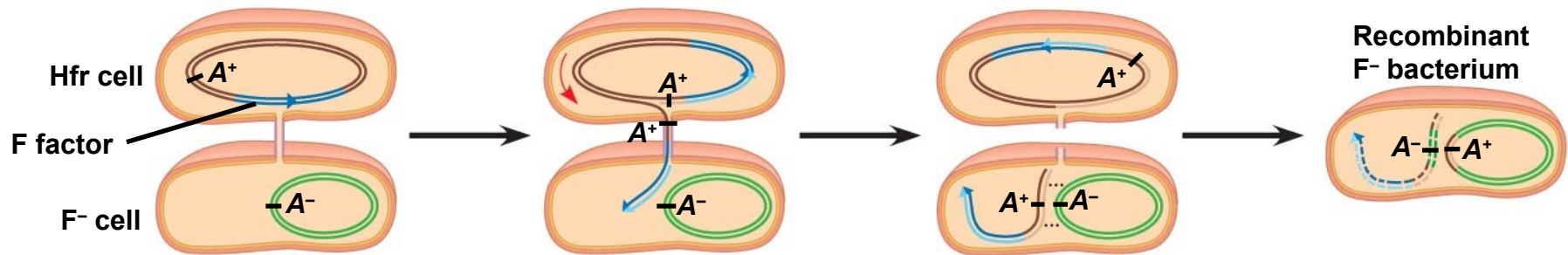
Sex pilus

1 μm

Conjugation



(a) Conjugation and transfer of an F plasmid



(b) Conjugation and transfer of part of an Hfr bacterial chromosome

R Plasmids and Antibiotic Resistance

- **R plasmids** carry genes for antibiotic resistance.
- Antibiotics select for bacteria with genes that are resistant to the antibiotics.
- Antibiotic resistant strains of bacteria are becoming more common.

Concept 27.3: Diverse nutritional and metabolic adaptations have evolved in prokaryotes

- **Phototrophs** obtain energy from light.
 - **Chemotrophs** obtain energy from chemicals.
 - **Autotrophs** require CO₂ as a carbon source.
 - **Heterotrophs** require an organic nutrient to make organic compounds.
 - These factors can be combined to give the four major modes of nutrition: photoautotrophy, chemoautotrophy, photoheterotrophy, and chemoheterotrophy.
-

Prokaryotic Nutritional Diversity

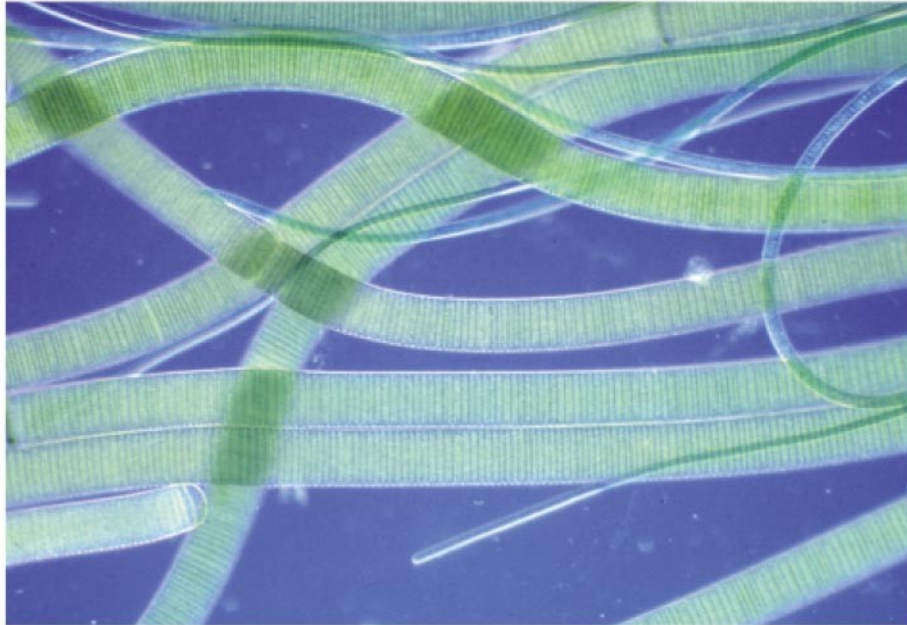
Table 27.1 Major Nutritional Modes

Mode of Nutrition	Energy Source	Carbon Source	Types of Organisms
Autotroph			
Photoautotroph	Light	CO ₂	Photosynthetic prokaryotes (for example, cyanobacteria); plants; certain protists (for example, algae)
Chemoautotroph	Inorganic chemicals	CO ₂	Certain prokaryotes (for example, <i>Sulfolobus</i>)
Heterotroph			
Photoheterotroph	Light	Organic compounds	Certain prokaryotes (for example, <i>Rhodobacter</i> , <i>Chloroflexus</i>)
Chemoheterotroph	Organic compounds	Organic compounds	Many prokaryotes (for example, <i>Clostridium</i>) and protists; fungi; animals; some plants

Cyanobacteria

- Cyanobacteria are *photoautotrophs* that generate O₂
- *Plant chloroplasts* likely evolved from cyanobacteria by *endosymbiosis*.

Cyanobacteria: One of the Major Groups of Bacteria



50 μm

**Two species of *Oscillatoria*,
filamentous cyanobacteria (LM)**

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

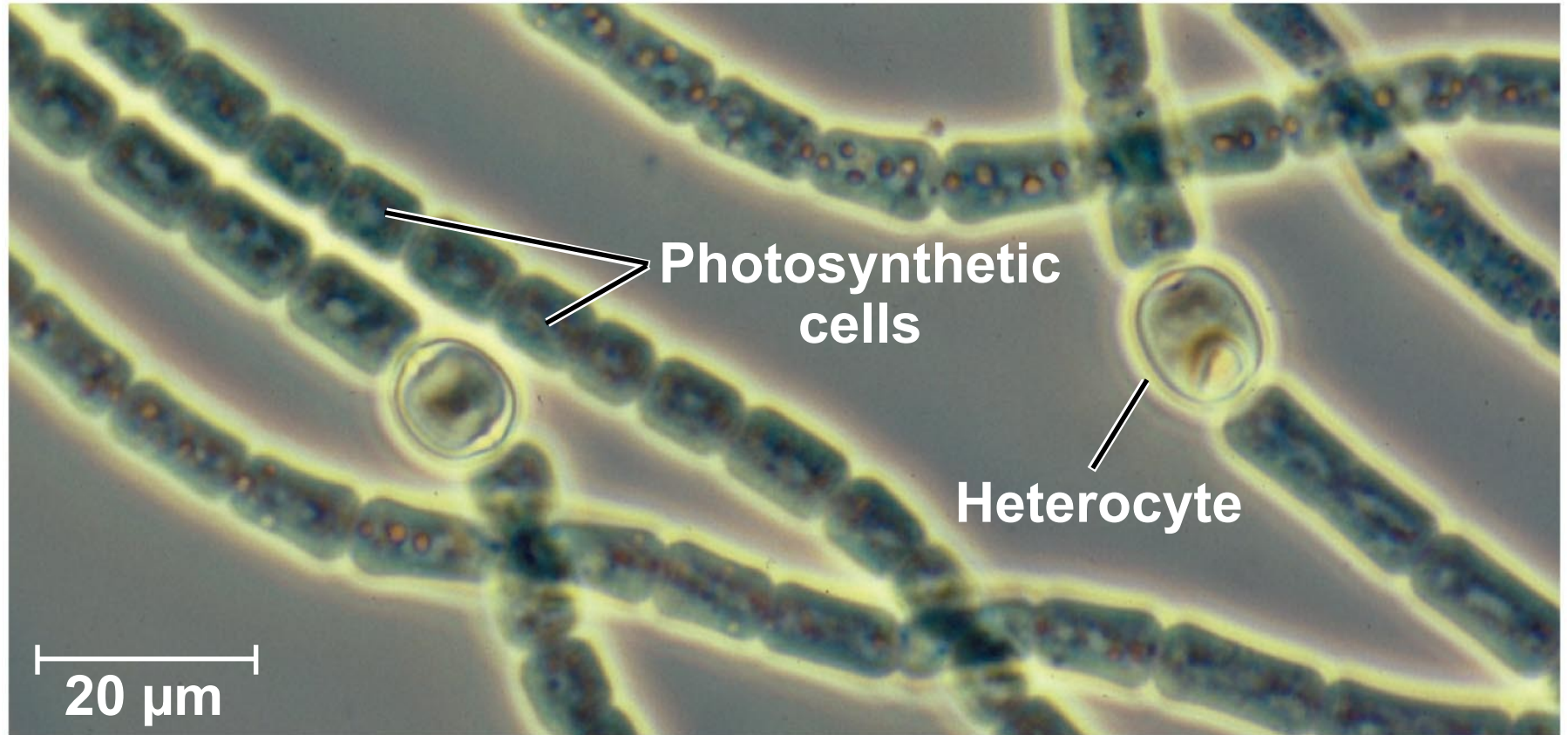
The Role of Oxygen in Metabolism

- *Prokaryotic metabolism varies* with respect to O_2 :
 - ***Obligate aerobes*** require O_2 for cellular respiration.
 - ***Obligate anaerobes*** are poisoned by O_2 and use fermentation or **anaerobic respiration**.
 - ***Facultative anaerobes*** can survive with or without O_2 .

Nitrogen Metabolism

- In *nitrogen fixation*, some prokaryotes convert atmospheric nitrogen (N_2) to ammonia (NH_3).
- *Metabolic Cooperation* between prokaryotes allows them to use environmental resources they could not use as individual cells.
- In the cyanobacterium *Anabaena*, photosynthetic cells and *nitrogen-fixing cells* called *heterocytes* exchange metabolic products.

Metabolic Cooperation between Bacteria Cells

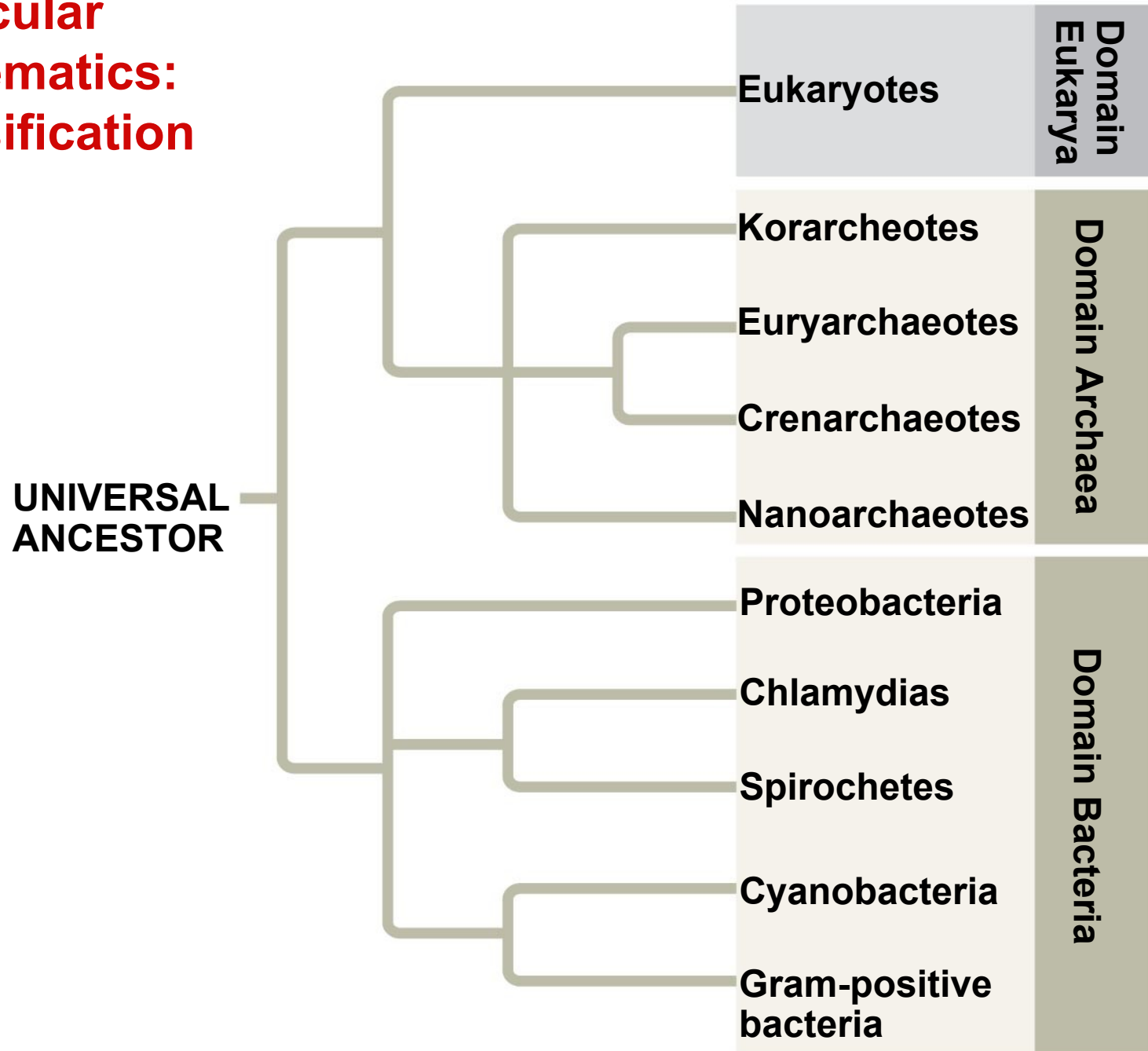


Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

Lessons from Molecular Systematics

- Molecular systematics is leading to a phylogenetic classification of prokaryotes.
- It allows systematists to identify major new clades.
- The use of polymerase chain reaction (PCR) has allowed for more rapid sequencing of prokaryote genomes.
- A handful of soil many contain 10,000 prokaryotic species.
- Horizontal gene transfer between prokaryotes obscures the root of the tree of life.

Molecular Systematics: Classification



Domain: Archaea

- **Archaea** are **prokaryotes** and share certain traits with bacteria and other traits with eukaryotes.
- Some archaea live in extreme environments and are called ***extremophiles***.
- ***Extreme halophiles*** live in highly saline, ***salty*** environments.
- ***Extreme thermophiles*** thrive in ***very hot*** environments.

Archaea appear to be more closely related to Eukarya than to Bacteria



Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

Table 27.2 A Comparison of the Three Domains of Life

CHARACTER	DOMAIN		
	Bacteria	Archaea	Eukarya
Nuclear envelope	Absent	Absent	Present
Membrane-enclosed organelles	Absent	Absent	Present
Peptidoglycan in cell wall	Present	Absent	Absent
Membrane lipids	Unbranched hydrocarbons	Some branched hydrocarbons	Unbranched hydrocarbons
RNA polymerase	One kind	Several kinds	Several kinds
Initiator amino acid for protein synthesis	Formyl-methionine	Methionine	Methionine
Introns in genes	Very rare	Present in some genes	Present
Response to the antibiotics streptomycin and chloramphenicol	Growth inhibited	Growth not inhibited	Growth not inhibited
Histones associated with DNA	Absent	Present in some species	Present
Circular chromosome	Present	Present	Absent
Growth at temperatures > 100°C	No	Some species	No

Extreme Thermophiles



-
- ***Methanogens*** live in swamps and marshes and produce methane as a waste product.
 - Methanogens are strict **anaerobes** and are poisoned by O₂
 - In recent years, genetic prospecting has revealed many new groups of archaea.
 - Some of these may offer clues to the early evolution of life on Earth.

Bacteria

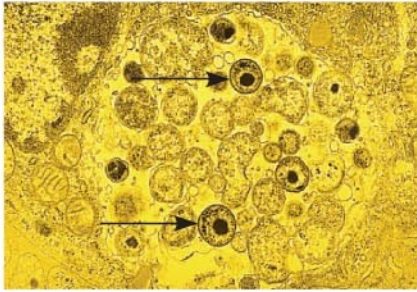
- Bacteria include the vast majority of prokaryotes of which most people are aware.
- Diverse nutritional types are scattered among the major groups of bacteria.
- *Rhizobium* are **nitrogen fixing bacteria**. They form root nodules in legumes (**mutualism ++**) and fix atmospheric N₂
- *Agrobacterium* produces tumors in plants and is used in genetic engineering.
- *Escherichia coli* resides in the intestines of many mammals and is not normally pathogenic.

Common Bacteria

- **Chlamydias** are parasitic bacteria that live within animal cells.
- *Chlamydia trachomatis* causes blindness and nongonococcal urethritis by sexual transmission.
- **Spirochetes** are helical heterotrophs.
- Some, such as *Treponema pallidum*, which causes syphilis, and *Borrelia burgdorferi*, which causes Lyme disease, are parasites.

Major Groups of Bacteria

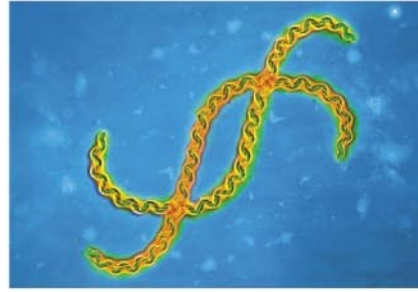
CHLAMYDIAS



2.5 μm

Chlamydia (arrows) inside an animal cell (colorized TEM)

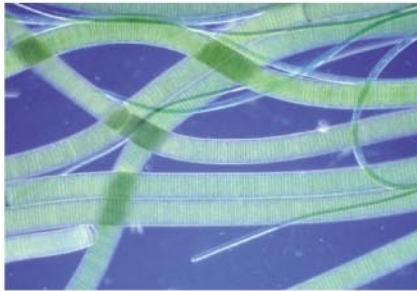
SPIROCHETES



5 μm

Leptospira, a spirochete (colorized TEM)

CYANOBACTERIA



50 μm

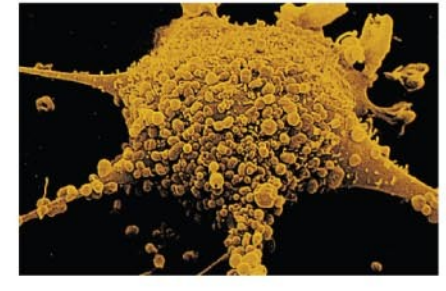
Two species of *Oscillatoria*, filamentous cyanobacteria (LM)

GRAM-POSITIVE BACTERIA



5 μm

Streptomyces, the source of many antibiotics (colorized SEM)



1 μm

Hundreds of mycoplasmas covering a human fibroblast cell (colorized SEM)

Gram-Positive Bacteria

- Gram-positive bacteria include
 - Actinomycetes, which decompose soil.
 - *Bacillus anthracis*, the cause of anthrax.
 - *Clostridium botulinum*, the cause of botulism.
 - Some *Staphylococcus* and *Streptococcus*, which can be pathogenic.
 - Mycoplasmas, the smallest known cells.

Prokaryotes play crucial roles in the biosphere.

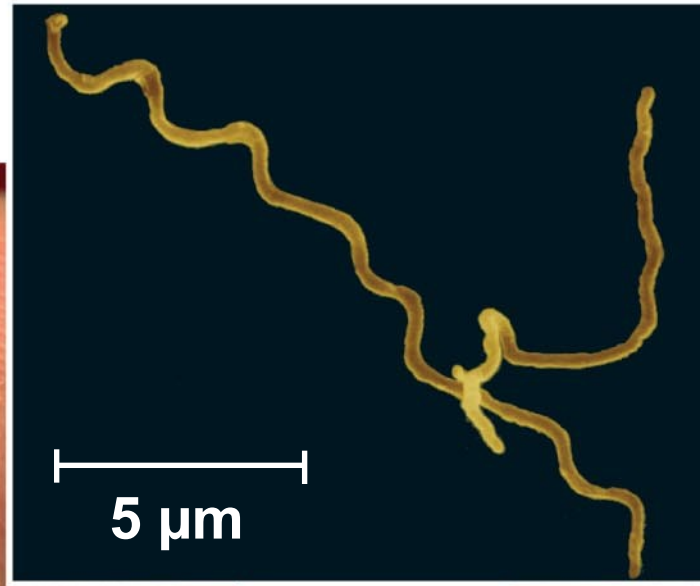
Chemical Cycling

- Prokaryotes play a major role in the recycling of chemical elements between the living and nonliving components of ecosystems.
 - *Chemoheterotrophic prokaryotes* function as **decomposers** (saphrophytes) breaking down corpses, dead vegetation, and waste products.
 - *Nitrogen-fixing prokaryotes* add usable nitrogen to the environment.
 - Prokaryotes can increase the availability of nitrogen, phosphorus, and potassium for plant growth.
 - Prokaryotes can also “immobilize” or decrease the availability of nutrients.
-

Ecological Interactions

- ***Symbiosis*** is an *ecological relationship in which two different species live in close contact*: a larger **host** and smaller **symbiont**.
- In ***mutualism***, **+ +** both symbiotic organisms benefit.
- In ***commensalism***, **+ 0** one organism benefits while neither harming nor helping the other.
- In ***parasitism***, **+ -** an organism called a **parasite** harms but does not kill its host. **Parasites** that **cause disease** are called **pathogens**.

Pathogens: Bacteria Causing Lyme Disease are transmitted by deer ticks



-
- *Pathogenic prokaryotes typically cause disease by releasing exotoxins or endotoxins.*
 - **Exotoxins** cause disease even if the prokaryotes that produce them are not present.
 - **Endotoxins** are released only when bacteria die and their cell walls break down.
 - Many pathogenic bacteria are potential weapons of bioterrorism.

Prokaryotes in Research and Technology

- Experiments using prokaryotes have led to important advances in **DNA technology**.
- Prokaryotes are the principal agents in **bioremediation**, the *use of organisms to remove pollutants from the environment*.
- Some other uses of prokaryotes:
 - Recovery of metals from ores
 - Synthesis of vitamins
 - Production of antibiotics, hormones, and other products.

Mutualism ++ Nitrogen Fixing Bacteria and Legume Plants

<i>Rhizobium</i> strain	Plant mass (g) after 12 weeks of growth
1	0.91
2	0.06
3	1.56
4	1.72
5	0.14
6	1.03

Source: J. J. Burdon, et al., Variation in the effectiveness of symbiotic associations between native rhizobia and temperate Australian *Acacia*: within species interactions, *Journal of Applied Ecology*, 36:398-408 (1999).

Note: In the absence of *Rhizobium*, after 12 weeks of growth, *Acacia* plants have a mass of about 0.1 g.

You should now be able to:

1. Distinguish between the cell walls of gram-positive and gram-negative bacteria.
2. State the function of the following features: capsule, fimbriae, sex pilus, nucleoid, plasmid, and endospore.
3. Explain how R plasmids confer antibiotic resistance on bacteria.
4. Explain the importance of / uses for prokaryotes.